1) a)

$$\begin{pmatrix}3 & -4\\ 1 & -1\end{pmatrix}\begin{pmatrix}2k+1 & -4k\\ k & 1-2k\end{pmatrix}=\begin{pmatrix}6k+3-4k & -12k-4+8k\\ 2k+1-k & -4k-1+2k\end{pmatrix}$$

$$=\begin{pmatrix}2k+3 & -4k-4\\ k+1 & -2k-1\end{pmatrix}$$

$$=\begin{pmatrix}2(k+1)+1 & -4(k+1)\\ k+1 & -2(k+1)+1\end{pmatrix}\text{ as req.}$$

1) b)

$$\begin{pmatrix}1 & 1\\ 0 & 2\end{pmatrix}\begin{pmatrix}1 & 2^k-1\\ 0 & 2^k\end{pmatrix}=\begin{pmatrix}1 & 2^k-1+2^k\\ 0 & 2\times 2k\end{pmatrix}$$

$$=\begin{pmatrix}1 & 2^1\times 2^k-1\\ 0 & 2^1\times 2k\end{pmatrix}$$

$$=\begin{pmatrix}1 & 2^{k+1}-1\\ 0 & 2^{k+1}\end{pmatrix}\text{ as req.}$$

1) c)

$$\begin{pmatrix}1 & 0\\ -1 & 5\end{pmatrix}\begin{pmatrix}1 & 0\\ -\frac{1}{4}(5^k-1) & 5^k\end{pmatrix}=\begin{pmatrix}1 & 0\\ -1-\frac{5}{4}(5^k-1) & 5\times 5^k\end{pmatrix}$$

$$=\begin{pmatrix}1 & 0\\ -1-\frac{5}{4}\times 5^k+\frac{5}{4} & 5^1\times 5^k\end{pmatrix}$$

$$=\begin{pmatrix}1 & 0\\ -\frac{1}{4}\times 5^1\times 5^k+\frac{1}{4} & 5^{k+1}\end{pmatrix}$$

$$=\begin{pmatrix}1 & 0\\ -\frac{1}{4}\times 5^{k+1}+\frac{1}{4} & 5^{k+1}\end{pmatrix}$$

$$=\begin{pmatrix}1 & 0\\ -\frac{1}{4}(5^{k+1}-1) & 5^{k+1}\end{pmatrix}\text{ as req.}$$

2)

Prove true for $n = 1$:

$\mathbf{A}^1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$

so true for $n = 1$

Assume true for $n = k$:

$\mathbf{A}^k = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$

Prove true for $n = k + 1$:

$\mathbf{A}^{k+1} = \mathbf{A}^1 \times \mathbf{A}^k = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} 1 & k+1 \\ 0 & 1 \end{pmatrix}$

which is $\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

3)

Prove true for $n = 1$:

$\mathbf{A}^1 = \begin{pmatrix} 1 & 2 \times 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$

so true for $n = 1$

Assume true for $n = k$:

$\mathbf{A}^k = \begin{pmatrix} 1 & 2k \\ 0 & 1 \end{pmatrix}$

Prove true for $n = k + 1$:

$\mathbf{A}^{k+1} = \mathbf{A}^1 \times \mathbf{A}^k = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 2k \\ 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} 1 & 2+2k \\ 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} 1 & 2(k+1) \\ 0 & 0 \end{pmatrix}$

which is $\begin{pmatrix} 1 & 2n \\ 0 & 1 \end{pmatrix}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

4)

Prove true for $n = 1$:

$\mathbf{A}^1 = \begin{pmatrix} 2+1 & -2 \\ 2 & 1-2 \end{pmatrix} = \begin{pmatrix} 3 & -2 \\ 2 & -1 \end{pmatrix}$

so true for $n = 1$

Assume true for $n = k$:

$\mathbf{A}^k = \begin{pmatrix} 2k+1 & -2k \\ 2k & 1-2k \end{pmatrix}$

Prove true for $n = k + 1$:

$\mathbf{A}^{k+1} = \mathbf{A}^1 \times \mathbf{A}^k = \begin{pmatrix} 3 & -2 \\ 2 & -1 \end{pmatrix}\begin{pmatrix} 2k+1 & -2k \\ 2k & 1-2k \end{pmatrix}$

$= \begin{pmatrix} 6k+3-4k & -6k-2+4k \\ 4k+2-2k & -4k-1+2k \end{pmatrix} = \begin{pmatrix} 2k-2 & -2k-2 \\ 2k+2 & -2k-1 \end{pmatrix}$

$= \begin{pmatrix} 2(k+1)+1 & -2(k+1) \\ 2(k+1) & 1-2(k+1) \end{pmatrix}$

which is $\begin{pmatrix} 2n+1 & -2n \\ 2n & 1-2n \end{pmatrix}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

5)

Prove true for $n = 1$:

$\mathbf{A}^1 = \begin{pmatrix} 2+1 & 2\mathrm{i} \\ 2\mathrm{i} & 1-2 \end{pmatrix} = \begin{pmatrix} 3 & 2\mathrm{i} \\ 2\mathrm{i} & -1 \end{pmatrix}$

so true for $n = 1$

Assume true for $n = k$:

$\mathbf{A}^k = \begin{pmatrix} 2k+1 & 2k\mathrm{i} \\ 2k\mathrm{i} & 1-2k \end{pmatrix}$

Prove true for $n = k + 1$:

$\mathbf{A}^{k+1} = \mathbf{A}^1 \times \mathbf{A}^k = \begin{pmatrix} 3 & 2\mathrm{i} \\ 2\mathrm{i} & -1 \end{pmatrix}\begin{pmatrix} 2k+1 & 2k\mathrm{i} \\ 2k\mathrm{i} & 1-2k \end{pmatrix}$

$= \begin{pmatrix} 6k+3-4k & 6k\mathrm{i}+2\mathrm{i}-4k\mathrm{i} \\ 4k\mathrm{i}+2\mathrm{i}-2k\mathrm{i} & -4k-1+2k \end{pmatrix} = \begin{pmatrix} 2k+3 & 2k\mathrm{i}+2\mathrm{i} \\ 2k\mathrm{i}+2\mathrm{i} & -1-2k \end{pmatrix}$

$= \begin{pmatrix} 2(k+1)+1 & 2(k+1)\mathrm{i} \\ 2(k+1)\mathrm{i} & 1-2(k+1) \end{pmatrix}$

which is $\begin{pmatrix} 2n+1 & 2n\mathrm{i} \\ 2n\mathrm{i} & 1-2n \end{pmatrix}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

6)

Prove true for $n = 1$:

$\mathbf{A}^1 = \begin{pmatrix} 2^1 & 4^1 - 2^1 \\ 0 & 4^1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 0 & 4 \end{pmatrix}$

so true for $n = 1$

Assume true for $n = k$:

$\mathbf{A}^k = \begin{pmatrix} 2^k & 4^k - 2^k \\ 0 & 4^k \end{pmatrix}$

Prove true for $n = k + 1$:

$\mathbf{A}^{k+1} = \mathbf{A}^1 \times \mathbf{A}^k = \begin{pmatrix} 2 & 2 \\ 0 & 4 \end{pmatrix}\begin{pmatrix} 2^k & 4^k - 2^k \\ 0 & 4^k \end{pmatrix}$

$= \begin{pmatrix} 2 \times 2^k & 2 \times 4^k - 2 \times 2^k + 2 \times 4^k \\ 0 & 4 \times 4^k \end{pmatrix}$

$= \begin{pmatrix} 2^1 \times 2^k & 4^1 \times 4^k - 2^1 \times 2^k \\ 0 & 4^1 \times 4^k \end{pmatrix}$

$= \begin{pmatrix} 2^{k+1} & 4^{k+1} - 2^{k+1} \\ 0 & 4^{k+1} \end{pmatrix}$

which is $\begin{pmatrix} 2^n & 4^n - 2^n \\ 0 & 4^n \end{pmatrix}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

7)

Prove true for $n = 1$:

$\mathbf{A}^1 = \begin{pmatrix} 2^1 & 1 \times 2^{1-1} \\ 0 & 2^1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$

so true for $n = 1$

Assume true for $n = k$:

$\mathbf{A}^k = \begin{pmatrix} 2^k & k \times 2^{k-1} \\ 0 & 2^k \end{pmatrix}$

Prove true for $n = k + 1$:

$\mathbf{A}^{k+1} = \mathbf{A}^1 \times \mathbf{A}^k = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 2^k & k \times 2^{k-1} \\ 0 & 2^k \end{pmatrix}$

$= \begin{pmatrix} 2 \times 2^k & 2k \times 2^{k-1} + 2^k \\ 0 & 2 \times 2^k \end{pmatrix}$

$= \begin{pmatrix} 2^1 \times 2^k & k \times 2^k + 2^k \\ 0 & 2^1 \times 2^k \end{pmatrix}$

$= \begin{pmatrix} 2^{k+1} & (k+1) \times 2^k \\ 0 & 2^{k+1} \end{pmatrix}$

$= \begin{pmatrix} 2^{k+1} & (k+1) \times 2^{(k+1)-1} \\ 0 & 2^{k+1} \end{pmatrix}$

which is $\begin{pmatrix} 2^n & n2^{n-1} \\ 0 & 2^n \end{pmatrix}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

8)

Prove true for $n = 1$:

$$\mathbf{A}^1 = \begin{pmatrix} 1 & \frac{1}{2}(3^1 - 1) \\ 0 & 3^1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix}$$

so true for $n = 1$

Assume true for $n = k$:

$$\mathbf{A}^k = \begin{pmatrix} 1 & \frac{1}{2}(3^k - 1) \\ 0 & 3^k \end{pmatrix}$$

Prove true for $n = k + 1$:

$$\mathbf{A}^{k+1} = \mathbf{A}^1 \times \mathbf{A}^k = \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 1 & \frac{1}{2}(3^k - 1) \\ 0 & 3^k \end{pmatrix}$$

$$= \begin{pmatrix} 1 & \frac{1}{2}(3^k - 1) + 3^k \\ 0 & 3 \times 3^k \end{pmatrix}$$

$$= \begin{pmatrix} 1 & \frac{1}{2}(3^k - 1 + 2 \times 3^k) \\ 0 & 3^1 \times 3^k \end{pmatrix}$$

$$= \begin{pmatrix} 1 & \frac{1}{2}(3 \times 3^k - 1) \\ 0 & 3^{k+1} \end{pmatrix}$$

$$= \begin{pmatrix} 1 & \frac{1}{2}(3^{k+1} - 1) \\ 0 & 3^{k+1} \end{pmatrix}$$

which is $\begin{pmatrix} 1 & \frac{1}{2}(3^n - 1) \\ 0 & 3^n \end{pmatrix}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

9)

Prove true for $n = 1$:

$$\mathbf{A}^1 = \begin{pmatrix} 1 & 0 \\ \frac{1}{3}(1-4^1) & 4^1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -1 & 4 \end{pmatrix}$$

so true for $n = 1$

Assume true for $n = k$:

$$\mathbf{A}^k = \begin{pmatrix} 1 & 0 \\ \frac{1}{3}(1-4^k) & 4^k \end{pmatrix}$$

Prove true for $n = k + 1$:

$$\mathbf{A}^{k+1} = \mathbf{A}^1 \times \mathbf{A}^k = \begin{pmatrix} 1 & 0 \\ -1 & 4 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ \frac{1}{3}(1-4^k) & 4^k \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ -1 + \frac{4}{3}(1-4^k) & 4 \times 4^k \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ -1 + \frac{4}{3} - \frac{1}{3} \times 4 \times 4^k & 4^1 \times 4^k \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ \frac{1}{3} - \frac{1}{3} \times 4^1 \times 4^k & 4^{k+1} \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ \frac{1}{3}(1-4^{k+1}) & 4^{k+1} \end{pmatrix}$$

which is $\begin{pmatrix} 1 & 0 \\ \frac{1}{3}(1-4^n) & 4^n \end{pmatrix}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

10) a)

$\mathbf{A}^2 = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 6 \\ 0 & 1 \end{pmatrix}$

$\mathbf{A}^3 = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 6 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 9 \\ 0 & 1 \end{pmatrix}$

$\mathbf{A}^4 = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 9 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 12 \\ 0 & 1 \end{pmatrix}$

10) b)

$\mathbf{A}^n = \begin{pmatrix} 1 & 3n \\ 0 & 1 \end{pmatrix}$

10) c)

Prove true for $n = 1$:

$\mathbf{A}^1 = \begin{pmatrix} 1 & 3 \times 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$

so true for $n = 1$

Assume true for $n = k$:

$\mathbf{A}^k = \begin{pmatrix} 1 & 3k \\ 0 & 1 \end{pmatrix}$

Prove true for $n = k + 1$:

$\mathbf{A}^{k+1} = \mathbf{A}^1 \times \mathbf{A}^k = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 3k \\ 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} 1 & 3k + 3 \\ 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} 1 & 3(k + 1) \\ 0 & 1 \end{pmatrix}$

which is $\begin{pmatrix} 1 & 3n \\ 0 & 1 \end{pmatrix}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

11)

Prove true for $n = 1$:

$$\mathbf{A}^1 = \begin{pmatrix} 4^1 & \frac{a}{3}(4^1 - 1) \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & a \\ 0 & 1 \end{pmatrix}$$

so true for $n = 1$

Assume true for $n = k$:

$$\mathbf{A}^k = \begin{pmatrix} 4^k & \frac{a}{3}(4^k - 1) \\ 0 & 1 \end{pmatrix}$$

Prove true for $n = k + 1$:

$$\mathbf{A}^{k+1} = \mathbf{A}^1 \times \mathbf{A}^k = \begin{pmatrix} 4 & a \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 4^k & \frac{a}{3}(4^k - 1) \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 4 \times 4^k & \frac{4}{3}a(4^k - 1) + a \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 4^1 \times 4^k & \frac{4}{3}a \times 4^k - \frac{4}{3}a + a \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 4^{k+1} & \frac{a}{3} \times 4 \times 4^k - \frac{a}{3} \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 4^{k+1} & \frac{a}{3}(4^{k+1} - 1) \\ 0 & 1 \end{pmatrix}$$

which is $\begin{pmatrix} 4^n & \frac{a}{3}(4^n - 1) \\ 0 & 1 \end{pmatrix}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

12) a)

Prove true for $n = 1$:

$\mathbf{A}^1 = \begin{pmatrix} 1-3 & -9 \\ 1 & 3+1 \end{pmatrix} = \begin{pmatrix} -2 & -9 \\ 1 & 4 \end{pmatrix}$

so true for $n = 1$

Assume true for $n = k$:

$\mathbf{A}^k = \begin{pmatrix} 1-3k & -9k \\ k & 3k+1 \end{pmatrix}$

Prove true for $n = k + 1$:

$\mathbf{A}^{k+1} = \mathbf{A}^1 \times \mathbf{A}^k = \begin{pmatrix} -2 & -9 \\ 1 & 4 \end{pmatrix}\begin{pmatrix} 1-3k & -9k \\ k & 3k+1 \end{pmatrix}$

$= \begin{pmatrix} -2+6k-9k & 18k-27k-9 \\ 1-3k+4k & -9k+12k+4 \end{pmatrix}$

$= \begin{pmatrix} -2-3k & -9k-9 \\ 1+k & 3k+4 \end{pmatrix}$

$= \begin{pmatrix} 1-3(k+1) & -9(k+1) \\ k+1 & 3(k+1)+1 \end{pmatrix}$

which is $\begin{pmatrix} 1-3n & -9n \\ n & 3n+1 \end{pmatrix}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

12) b)

$\begin{pmatrix} -2 & -9 \\ 1 & 4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$

$-2x - 9y = x \Rightarrow -9y = 3x \Rightarrow y = -\frac{1}{3}x$

$x + 4y = y \Rightarrow 3y = -x \Rightarrow y = -\frac{1}{3}x$

So any point on the line $y = -\frac{1}{3}x$ is invariant.

12) c)

$\begin{pmatrix} 1-3n & -9n \\ n & 3n+1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$

$(1-3n)x - 9ny = x \Rightarrow -9ny = 3nx \Rightarrow y = -\frac{1}{3}x$

$nx + 3ny + y = y \Rightarrow 3ny = -nx \Rightarrow y = -\frac{1}{3}x$

So any point on the line $y = -\frac{1}{3}x$ is invariant.

13) a)

$\det \mathbf{A} = 4 + a^2 = 0$

So $a = \pm 2\mathrm{i}$

13) b)

$\mathbf{A} = \begin{pmatrix} 2 & -2\mathrm{i} \\ 2\mathrm{i} & 2 \end{pmatrix}$ or $\mathbf{A} = \begin{pmatrix} 2 & 2\mathrm{i} \\ -2\mathrm{i} & 2 \end{pmatrix}$

Prove true for $n = 1$:

$\mathbf{A}^1 = 4^{1-1}\mathbf{A} = 4^0\mathbf{A} = \mathbf{A}$

so true for $n = 1$

Assume true for $n = k$:

$\mathbf{A}^n = 4^{k-1}\mathbf{A}$

Prove true for $n = k + 1$:

If $a = -2\mathrm{i}$

$\mathbf{A}^{k+1} = \mathbf{A}^1 \times \mathbf{A}^k = \begin{pmatrix} 2 & -2\mathrm{i} \\ 2\mathrm{i} & 2 \end{pmatrix} \times 4^{k-1} \begin{pmatrix} 2 & -2\mathrm{i} \\ 2\mathrm{i} & 2 \end{pmatrix}$

$= 4^{k-1} \begin{pmatrix} 4+4 & -4\mathrm{i}-4\mathrm{i} \\ 4\mathrm{i}+4\mathrm{i} & 4+4 \end{pmatrix} = 4^{k-1} \begin{pmatrix} 8 & -8\mathrm{i} \\ 8\mathrm{i} & 8 \end{pmatrix}$

$= 4 \times 4^{k-1} \begin{pmatrix} 2 & -2\mathrm{i} \\ 2\mathrm{i} & 2 \end{pmatrix}$

$= 4^1 \times 4^{k-1}\mathbf{A}$

$= 4^k\mathbf{A}$

which is $4^{n-1}\mathbf{A}$ with $k + 1$ substituted in.

If $a = 2\mathrm{i}$

$\mathbf{A}^{k+1} = \mathbf{A}^1 \times \mathbf{A}^k = \begin{pmatrix} 2 & 2\mathrm{i} \\ -2\mathrm{i} & 2 \end{pmatrix} \times 4^{k-1} \begin{pmatrix} 2 & 2\mathrm{i} \\ -2\mathrm{i} & 2 \end{pmatrix}$

$= 4^{k-1} \begin{pmatrix} 4+4 & 4\mathrm{i}+4\mathrm{i} \\ -4\mathrm{i}-4\mathrm{i} & 4+4 \end{pmatrix} = 4^{k-1} \begin{pmatrix} 8 & 8\mathrm{i} \\ -8\mathrm{i} & 8 \end{pmatrix}$

$= 4 \times 4^{k-1} \begin{pmatrix} 2 & 2\mathrm{i} \\ -2\mathrm{i} & 2 \end{pmatrix}$

$= 4^1 \times 4^{k-1}\mathbf{A}$

$= 4^k\mathbf{A}$

which is $4^{n-1}\mathbf{A}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

14)

Prove true for $n = 1$:

$\mathbf{B}^1\mathbf{A}\mathbf{B}^1 = \mathbf{B}(\mathbf{A}\mathbf{B}) = \mathbf{B}(\mathbf{B}^4\mathbf{A}) = \mathbf{B}^5\mathbf{A} = \mathbf{I}\mathbf{A} = \mathbf{A}$

so true for $n = 1$

Assume true for $n = k$:

$\mathbf{B}^k\mathbf{A}\mathbf{B}^k = \mathbf{A}$

Prove true for $n = k + 1$:

$\mathbf{B}^{k+1}\mathbf{A}\mathbf{B}^{k+1} = \mathbf{B}\mathbf{B}^k\mathbf{A}\mathbf{B}^k\mathbf{B}$

$= \mathbf{B}(\mathbf{B}^k\mathbf{A}\mathbf{B}^k)\mathbf{B}$

$= \mathbf{B}\mathbf{A}\mathbf{B}$

$= \mathbf{B}(\mathbf{B}^4\mathbf{A})$

$= \mathbf{B}^5\mathbf{A}$

$= \mathbf{I}\mathbf{A}$

$= \mathbf{A}$

So true for $n = k + 1$

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

15) a)

Prove true for $n = 1$:

$$\mathbf{A}^1 = \begin{pmatrix} 1 & 1 & \frac{1}{2}(1+1) \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

so true for $n = 1$

Assume true for $n = k$:

$$\mathbf{A}^k = \begin{pmatrix} 1 & k & \frac{1}{2}k(k+1) \\ 0 & 1 & k \\ 0 & 0 & 1 \end{pmatrix}$$

Prove true for $n = k + 1$:

$$\mathbf{A}^{k+1} = \mathbf{A}^1 \times \mathbf{A}^k = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & k & \frac{1}{2}k(k+1) \\ 0 & 1 & k \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & k+1 & \frac{1}{2}k(k+1)+k+1 \\ 0 & 1 & k+1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & k+1 & \frac{1}{2}(k(k+1)+2k+2) \\ 0 & 1 & k+1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & k+1 & \frac{1}{2}(k+1)(k+2) \\ 0 & 1 & n \\ 0 & 0 & 1 \end{pmatrix}$$

which is $\begin{pmatrix} 1 & n & \frac{1}{2}n(n+1) \\ 0 & 1 & n \\ 0 & 0 & 1 \end{pmatrix}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

15) b)

Prove true for $n = 1$:

$$\mathbf{A}^1 = \begin{pmatrix} 1 & 1 & \frac{1}{2}(1+3) \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

so true for $n = 1$

Assume true for $n = k$:

$$\mathbf{A}^k = \begin{pmatrix} 1 & k & \frac{1}{2}k(k+3) \\ 0 & 1 & k \\ 0 & 0 & 1 \end{pmatrix}$$

Prove true for $n = k + 1$:

$$\mathbf{A}^{k+1} = \mathbf{A}^1 \times \mathbf{A}^k = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & k & \frac{1}{2}k(k+3) \\ 0 & 1 & k \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & k+1 & \frac{1}{2}k(k+3)+k+2 \\ 0 & 1 & k+1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & k+1 & \frac{1}{2}(k(k+3)+2k+4) \\ 0 & 1 & k+1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & k+1 & \frac{1}{2}(k^2+3k+2k+4) \\ 0 & 1 & k+1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & k+1 & \frac{1}{2}(k^2+5k+4) \\ 0 & 1 & k+1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & k+1 & \frac{1}{2}(k+1)(k+4) \\ 0 & 1 & k+1 \\ 0 & 0 & 1 \end{pmatrix}$$

which is $\begin{pmatrix} 1 & n & \frac{1}{2}n(n+3) \\ 0 & 1 & n \\ 0 & 0 & 1 \end{pmatrix}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

15) c)

Prove true for $n = 1$:

$$\mathbf{A}^1 = \begin{pmatrix} 2^1 & 1-2^1 & 3(2^1-1) \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -1 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

so true for $n = 1$

Assume true for $n = k$:

$$\mathbf{A}^k = \begin{pmatrix} 2^k & 1-2^k & 3(2^k-1) \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Prove true for $n = k + 1$:

$$\mathbf{A}^{k+1} = \mathbf{A}^1 \times \mathbf{A}^k = \begin{pmatrix} 2 & -1 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 2^k & 1-2^k & 3(2^k-1) \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 2 \times 2^k & 2(1-2^k)-1 & 6(2^k-1)+3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 2^1 \times 2^k & 2-2\times 2^k-1 & 3(2\times 2^k-2)+3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 2^{k+1} & 1-2^1\times 2^k & 3(2^1\times 2^k-2+1) \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 2^{k+1} & 1-2^{k+1} & 3(2^{k+1}-1) \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

which is $\begin{pmatrix} 2^n & 1-2^n & 3(2^n-1) \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$